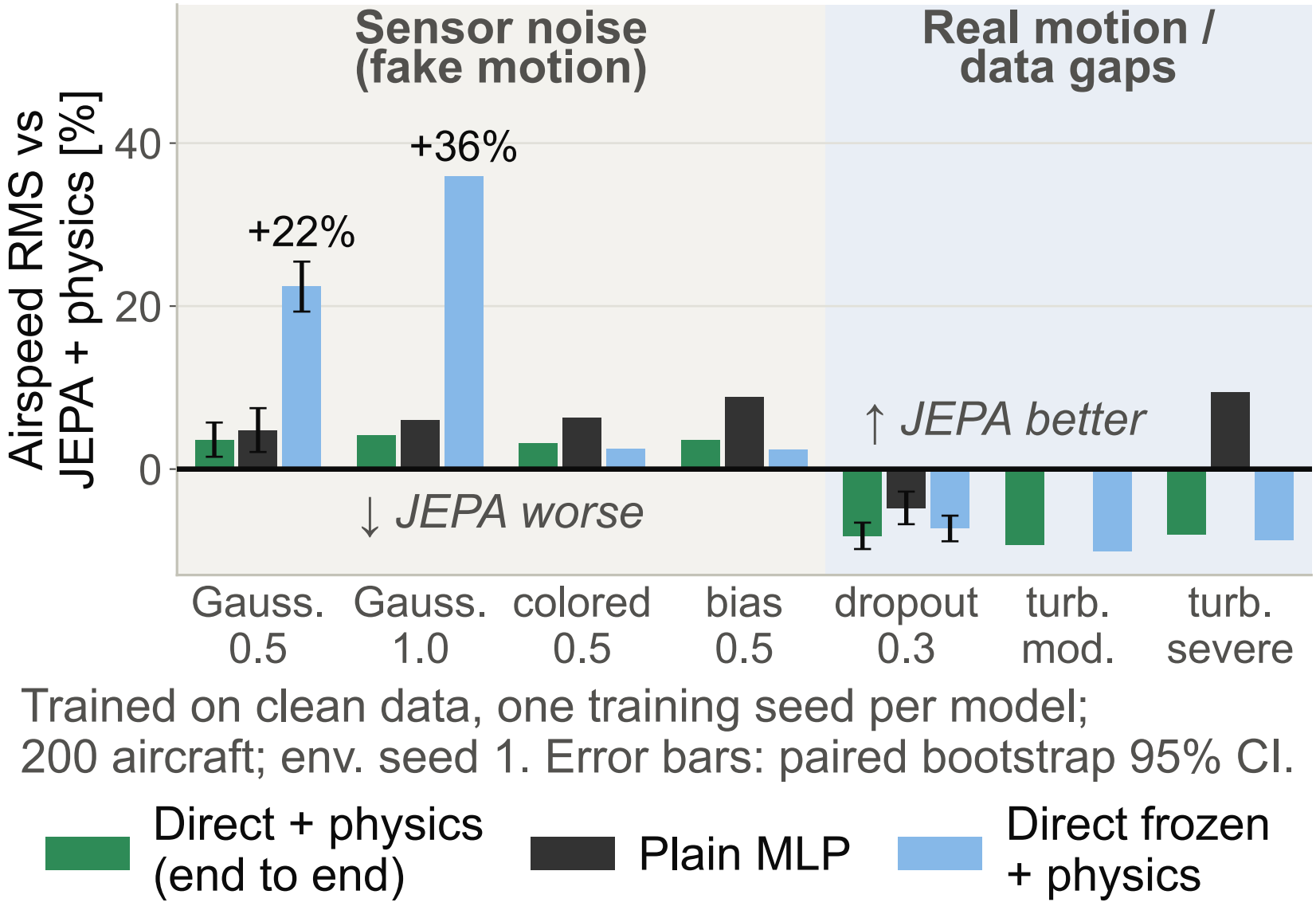

Sensor noise
(fake motion)
Real motion /
data gaps
Airspeed RMS vs
JEPA + physics [%]
40
20
0
+22%
+36%
↑ JEPA better
↓ JEPA worse
Gauss.
0.5
Gauss.
1.0
colored
0.5
bias
0.5
dropout
0.3
turb.
mod.
turb.
severe
Trained on clean data, one training seed per model;
200 aircraft; env. seed 1. Error bars: paired bootstrap 95% CI.
Direct + physics
(end to end)
Plain MLP
Direct frozen
+ physics